

Executive progress report

From one sentence to a CDR-ready design: systems engineering with Claude as the engineering team

Document	BOATY-PR-001
Issue	Issue 1, 1 October 2026
Audience	Executive stakeholders
Status	Briefing; summarises the baselined set (tag sdr-baseline-1) and the detailed-design drafts
Evidence	software results 2026-09-30; CAD build c856b2a; electrical build 82541bc

Executive summary

The brief was one sentence from a father and his four-year-old son: explore the pond, take pictures of the ducks, come back, with one hard rule: **we cannot lose it**. Behind that sentence sits a real autonomous system (a boat, its autopilot, a mission computer, a bank-side Mission Control and a natural-language planner using Claude) with real safety, cost and mass constraints.

We ran it as a formal systems-engineering programme in which Claude did the engineering work (requirements, architecture, software, simulation, CAD, schematics, analysis and every document) and the owner made the decisions at reviews. Three days after the first sentence the project has passed its System Design Review, has a baselined document set, has 83 simulated missions flown on the real autopilot firmware, and has a full mechanical and electrical detailed design ready for the Critical Design Review.

184 system requirements (137 Must)	247 derived subsystem requirements	16 baselined documents, all built from code	14 change requests raised and tracked
80/83 simulated missions pass on real autopilot firmware	0 simulation failures	68 design checks pass (mechanical + electrical)	3 days from one sentence to CDR-ready

Where we are

Phase	Status	Evidence
Concept selection	Done	7 options scored; modular GPS catamaran on ArduPilot chosen (4.45 / 5)
Requirements & architecture	Done	SRS 184 requirements, ADD, ICD, 8 subsystem specs, FMEA, budgets
System Design Review (SDR)	GO, baselined	15 review findings (RIDs) raised and closed; baseline tagged sdr-baseline-1
Software & simulation	Done	83 scenarios: 80 pass, 3 known gaps recorded as findings, 0 fail; 308 unit tests
Mechanical detailed design	Draft	CadQuery model; 33 checks pass, 3 to test, 4 open (CR-09 to CR-11)
Electrical detailed design	Draft	Netlist, 6 schematic sheets, 3 boards; 35 pass, 4 to test, 2 open (CR-12 to CR-14)
Critical Design Review (CDR)	Next	Decide CR-09 to CR-14, then build

What needs a decision. The design is honest about where it misses: the boat is **2165 g** against an 1800 g target (CR-09), and the bill of materials is **£229 against the £185 cap** (CR-14). Both are framed as owner decisions for the CDR with options and consequences, not hidden. Neither affects safety: the boat still floats, stays stable and passes every hydrostatic check at the heavier mass.

1. How we work: the V, with Claude as the engineering team

The programme follows the classic systems-engineering V. Down the left side the need is decomposed into requirements, architecture, subsystem specifications and detailed design; up the right side the product is built, integrated and verified against each level. Review gates stop work moving on until the level above is sound.

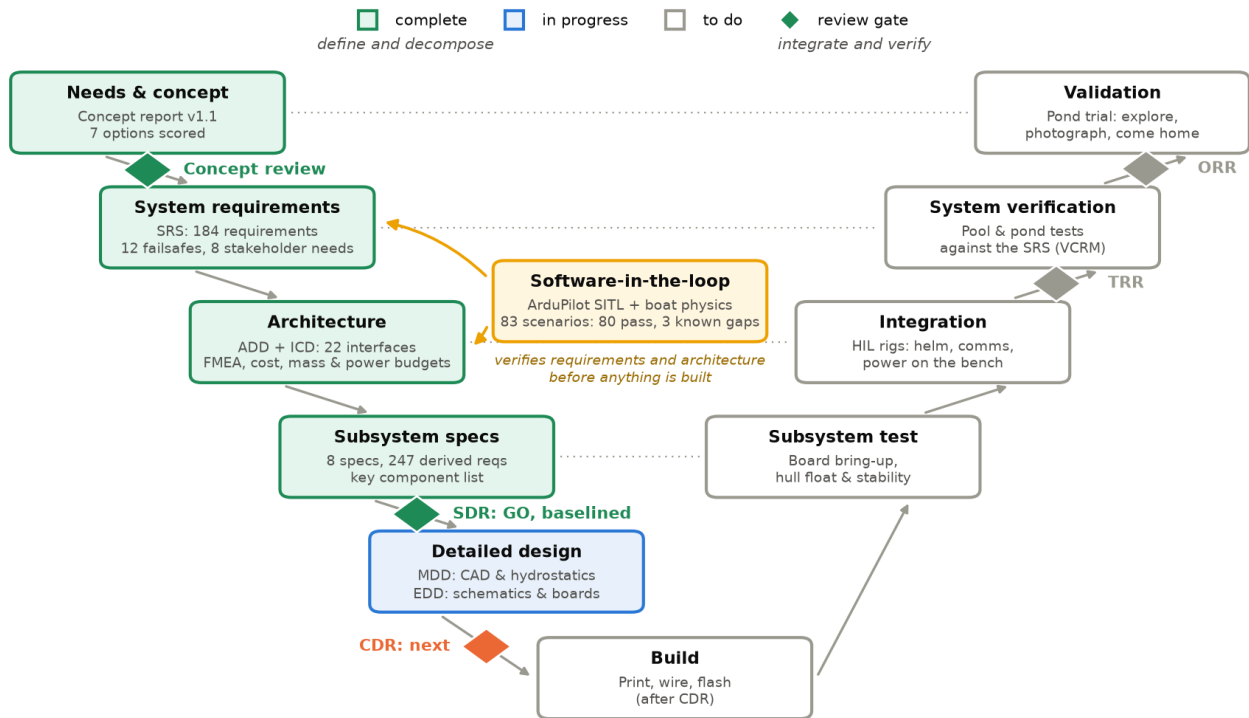


Figure 1. Progress on the V. Green is complete, blue is in progress. The yellow box is the difference: a software-in-the-loop model of the whole system, flown on the real autopilot firmware, lets us verify requirements and architecture before anything is built.

The prompting pattern

The approach is not "ask the AI to design a boat". It is the same discipline we would ask of a human team, written into how Claude is directed:

- **One step of the V at a time, each with a defined deliverable.** The owner asks for a concept report, then a requirements specification, then an architecture, and so on. Each is a controlled PDF with an issue, a revision history and traceability to the level above.
- **Documents are built from code, not typed.** Requirements live in Python data; the PDF, the traceability matrix and the checks are generated from them. The CAD model writes its own mass and hydrostatics results; the netlist writes its own schematics and rule check. A number in a document is always the number the analysis produced.
- **Claude proposes, the owner decides.** Every real choice (budget, scope, the wildlife rule, the link-loss timing) was put to the owner as a decision with options and recorded. Fourteen change requests (CR-01 to CR-14) carry those decisions into the documents.
- **Independent review, with findings.** At the SDR, Claude reviewed its own baseline as a sceptical reviewer would and raised 15 findings (9 Major), including a real safety gap: return-to-home could cut straight across a no-go zone. All 15 were closed with evidence before the owner said GO.
- **Baseline and change control.** The SDR documents are frozen at git tag **sdr-baseline-1**. The detailed design may not edit them; where it disagrees, it raises a numbered change request instead.

- **Honesty is a requirement.** Known gaps are reported as known gaps, failed checks as failures with a CR, and nothing is rounded into a pass.

Three days, start to CDR-ready

Day	Milestone
28 Sep	Concept selection report; SRS Issue A; owner decisions; architecture (ADD); interface control document; subsystem specifications; FMEA; operations manual; key component list; simulator slice 1 (autopilot in the loop)
29 Sep	Simulator slices 2-3: the boat services on a simulated Pi Zero, then Mission Control end to end on ArduPilot Rover SITL with a catamaran physics model
30 Sep	Live Claude planner evaluation 39/39; System Design Review: 15 RIDs raised, five work packages to close them, full regression, owner GO, baseline tagged
30 Sep	Mechanical detailed design (CadQuery) and electrical detailed design (netlist to schematics, boards and harness), each with a design description

2. Concept selection

Seven genuinely different ways to meet the brief were generated and scored against four weighted criteria: **reliability (30%)**, because 'we cannot lose it' is the one hard rule; **cost (25%)**; **fun to build with a four-year-old (25%)**; and **intelligence and learning (20%)**. Options ranged from buying a GPS bait boat to a scratch-built autopilot.

	Option	Score	Why it lost (or won)
G	Modular GPS catamaran	4.45	Proven failsafes & geofence, fully modular build, quiet, stable camera platform, clear upgrade path.
D	ArduPilot airboat	3.70	Loud (scares the ducks we want to photograph), blown about by wind, exposed spinning props near small fingers.
C	Scratch-built autopilot	3.65	Every failsafe is our own untested code - the riskiest way to meet 'never lose it'.
E	Paddle-wheel explorer	3.50	Slow, heavy, big to print; poor into wind; gearbox wear.
F	Converted RC toy boat	3.20	Tiny, not sealed, too fast, no room for camera/batteries; nothing modular to build.
B	Tethered boat	3.15	Tether snags reeds/weed and ducks; 'explore' limited to a circle; little autonomy to learn.
A	Hacked GPS bait boat	2.85	£150-300 before mods; closed firmware; little for your son to build.

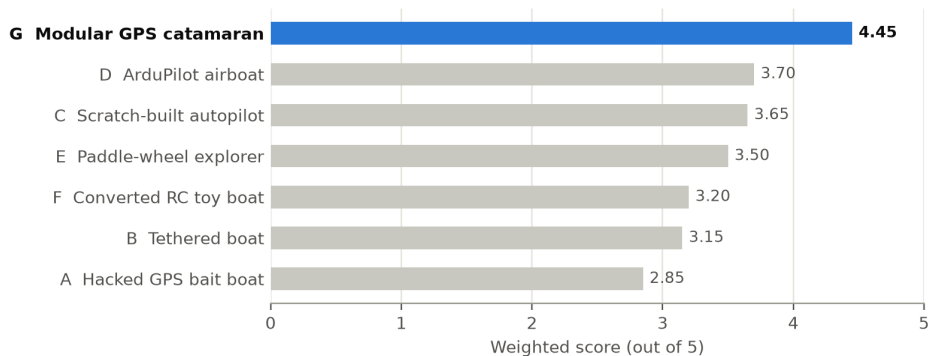


Figure 2. Weighted scores.

The winner, a **modular 3D-printed catamaran on ArduPilot**, wins because it gets proven safety functions (geofence, failsafes, return-to-launch) from mature open-source autopilot firmware instead of writing them, while keeping everything else buildable and playful. The scratch-built option was the most fun to code but scored worst on the hard rule: every failsafe would be untested home-made code.

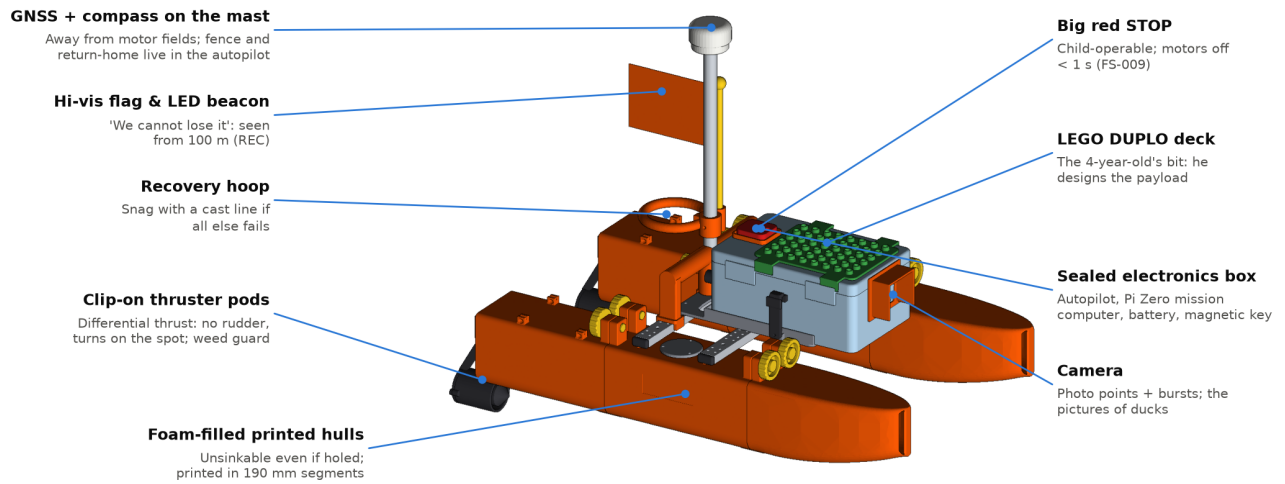


Figure 3. The chosen concept, as it now exists in the detailed CAD model. Every callout traces to a requirement in the SRS.

3. Requirements and architecture

How the requirements were developed

The one-line brief was expanded into 8 stakeholder needs (the child, the parent, the pond and the people and wildlife on it), then into 184 system requirements in 22 areas, 137 of them Must. Each has a verification method (test, demonstration, analysis or inspection) and a stage (simulation, bench, pool or pond), and 12 named failsafes define exactly what the boat does when something goes wrong. The architecture then allocated every requirement to one of eight subsystems, which have their own specifications (247 derived requirements), an interface control document with 22 interfaces, a design FMEA and mass, power and cost budgets.

The tensions, and how they were resolved

The value of the process is in the tensions it surfaces early, while they are cheap to resolve:

Tension	Resolution
Use AI to plan missions vs a child's boat must be safe	Claude only proposes. A deterministic validator checks every plan against the fence and no-go zones; the autopilot enforces the fence and failsafes independently of all Python and AI. Safety never depends on the language model.
Never lose it vs cost	Budget rose from £100 to £160 (CR-01) and £185 (CR-03) by owner decision, because reliability features (autopilot, GNSS, foam-filled hulls, hi-vis) are the ones not to cut. Detailed design now shows £229: CR-14.
Take pictures of the ducks vs don't disturb wildlife	Found at the SDR (RID-04). CR-07 reworded the rule: photograph from a stand-off, never chase; nesting sites are 15 m exclusions the site linter enforces.
A four-year-old operates it vs adult control	Child gets four big buttons (GO, COME HOME, STOP, TALK); arming, approval and edits need an adult PIN. STOP always works, in under 1 s.
Python for the family's software vs autopilot firmware is C++	A Pi Zero mission computer runs the Python boat services alongside the unmodified ArduPilot helm, resolving SWE-001.
Small, light, printable vs robust and stable	Hull segments under 200 mm, foam-filled; the detailed design shows the mass target is not met (CR-09) while stability has large margins.

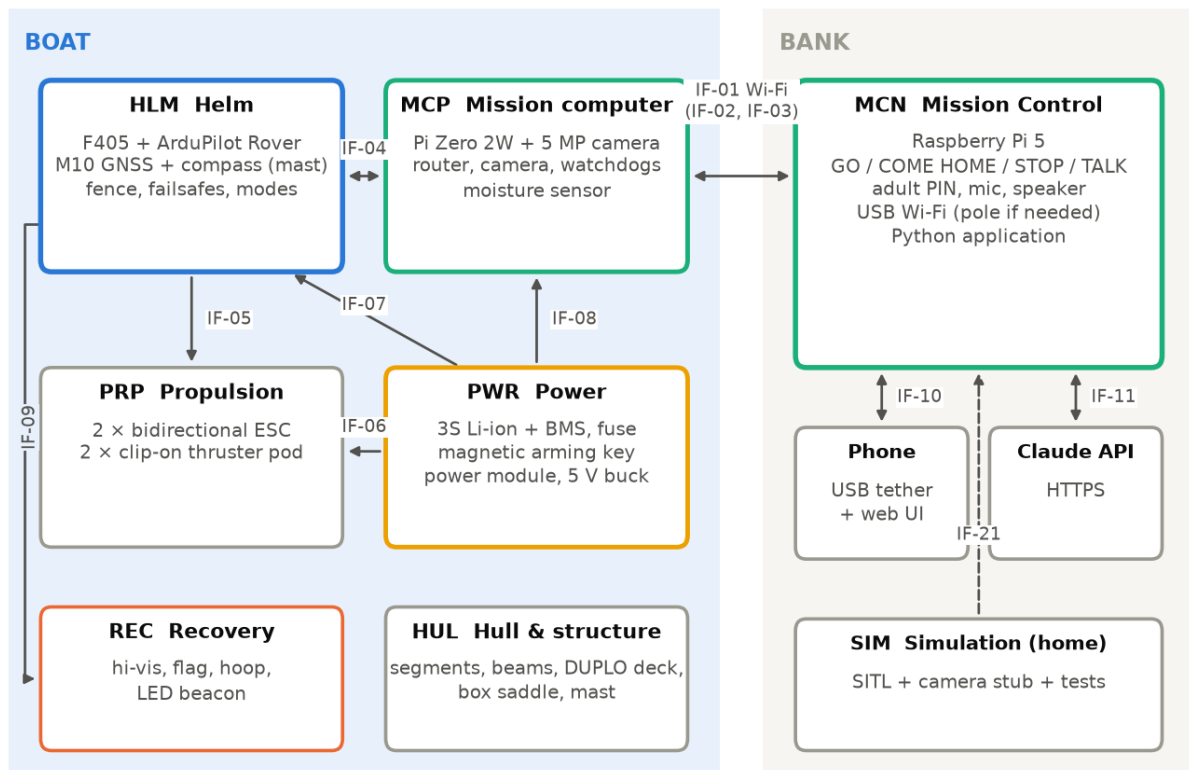


Figure 4. High-level architecture: six boat subsystems, Mission Control on the bank, and the simulation subsystem that stands in for the boat in testing. IF-xx are the controlled interfaces.

4. The system model and software-in-the-loop testing

The system model is not a diagram: it is the real software, flown against simulated physics. ArduPilot Rover 4.7.1 firmware (pinned to a specific commit) runs in software-in-the-loop (SITL) against a catamaran physics model with two thrusters, wind, weed drag and sensor faults. The Pi Zero boat services, Mission Control, the child's button panel, the photo pipeline and the Claude planner all run as they will on the hardware. Each test is a **mission-driven scenario** traced to the requirements it verifies: a mission is planned, approved, armed and flown, a fault is injected, and the test judges the true position of the simulated boat, not what the boat believes.

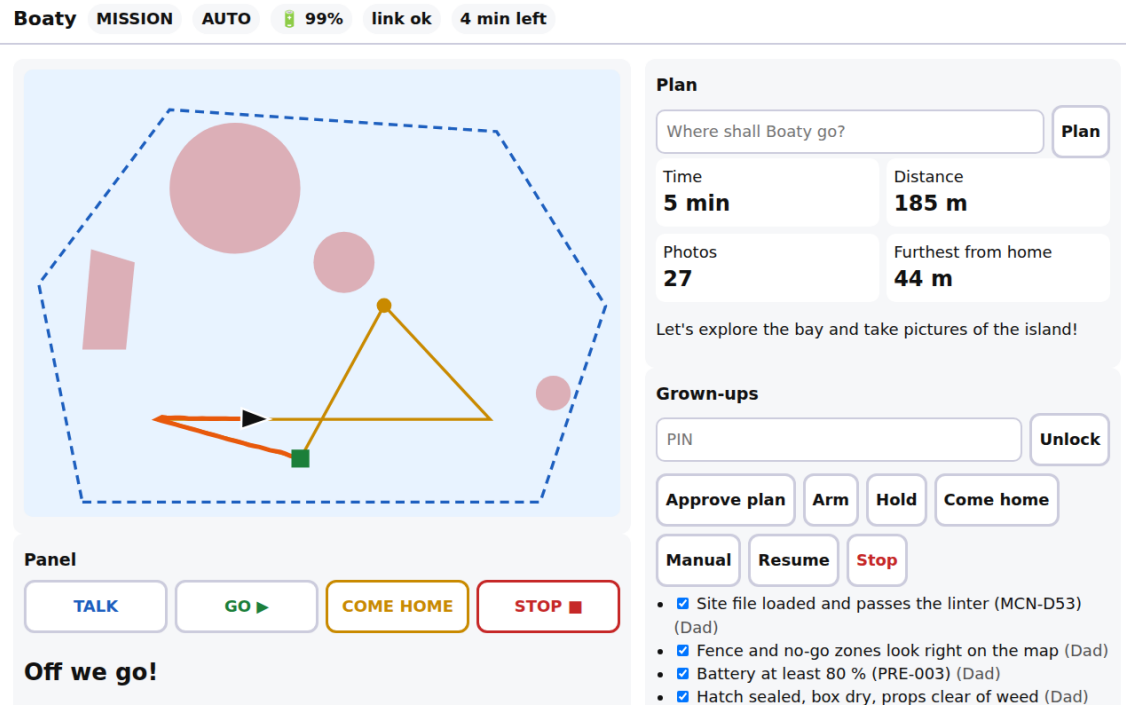


Figure 5. Screenshot of Mission Control's web page during the end-to-end test, served live from the simulation and captured in a headless browser. The boat (black arrow) is flying the plan (gold) and its track so far is drawn in orange; the no-go zones are pink and the fence is dashed blue. The child's four buttons are bottom left; everything on the right needs an adult PIN.

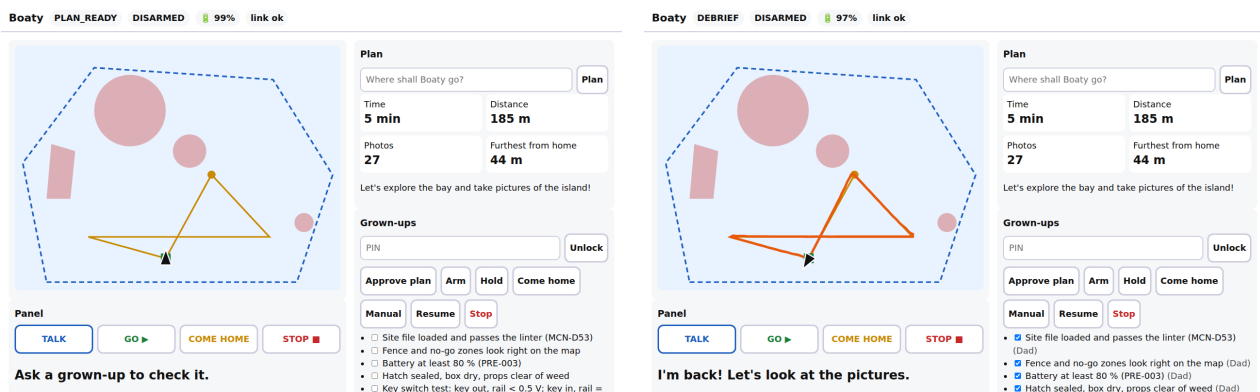


Figure 6. Before and after. Left: the plan Claude proposed from the typed instruction, validated and waiting for an adult to approve and sign the checklist. Right: home, disarmed, and the boat says so.

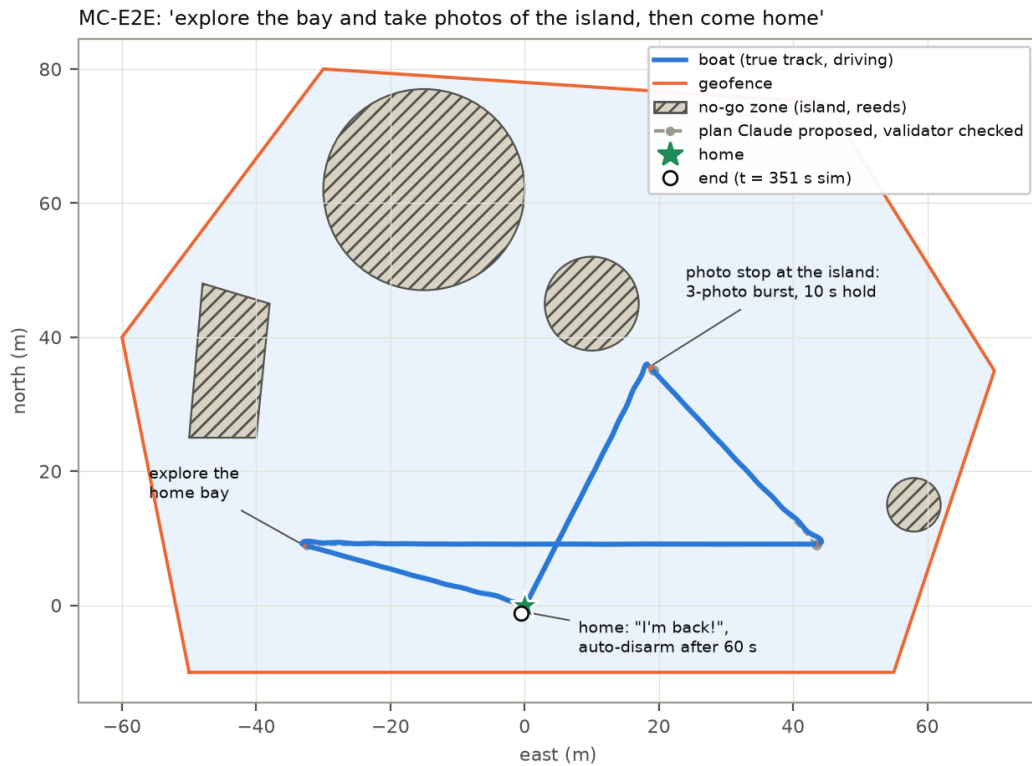


Figure 7. The nominal mission (MC-E2E). A typed instruction becomes a plan (Claude), is validated against the fence and no-go zones, read back from the autopilot and checksum-matched, flown, photographed, and brought home, where the boat disarms itself after 60 s and the photos are synced and hash-verified. The blue line is the simulated boat's true track.

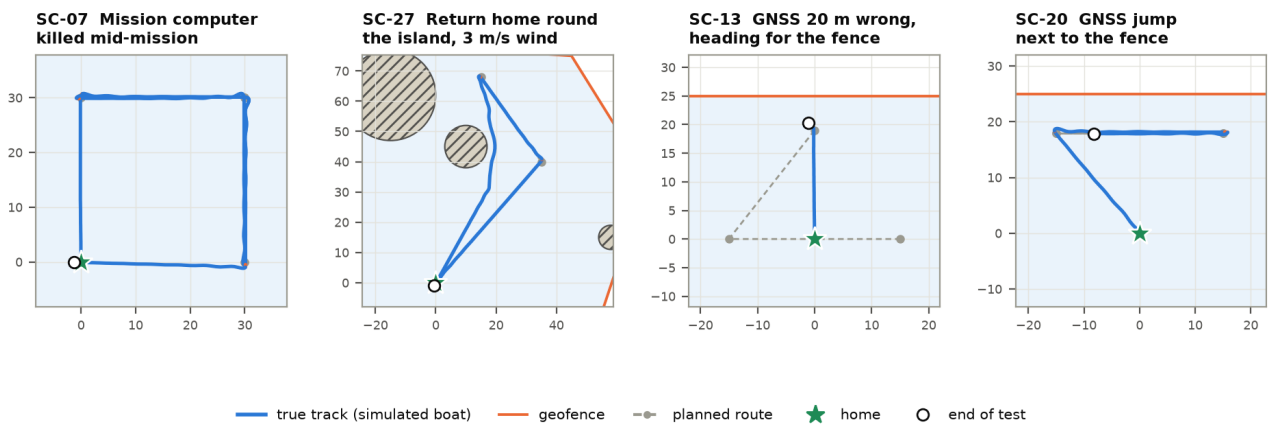


Figure 8. Edge cases, each flown on the real firmware. SC-07: the mission computer dies and the autopilot alone completes the mission. SC-27: return-to-home from behind the island, in a 3 m/s cross-wind, goes round the no-go zone (the SDR safety finding, now fixed and proven). SC-13: the satellite position is suddenly 20 m wrong while heading for the fence, and the boat never leaves it. SC-20: a 50 m GNSS jump for 2 s beside the fence line; the boat stays inside.

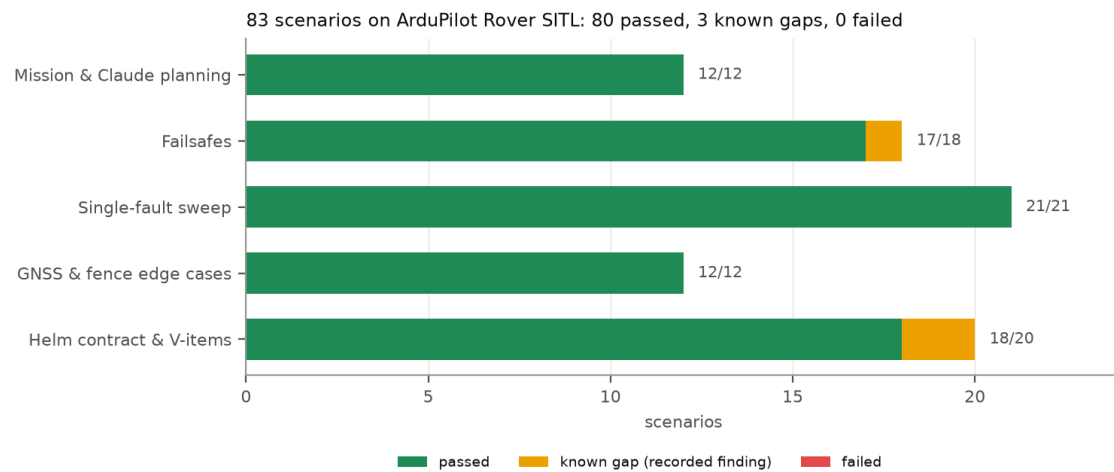


Figure 9. Results of the full simulation regression.

80 of 83 pass, 0 fail. The 3 known gaps are places where the stock autopilot on its own does not meet a requirement: its crash check is too slow for the 5 s stuck rule (V-05), GUIDED mode decelerates rather than cutting the motors at its 3 s timeout (V-11), and it does not stop the motors within 3 s of losing GNSS (SC-04, autopilot alone). Each is a recorded finding with a design response; for stuck and GNSS loss the boat services cover it, and those scenarios (SC-04, SC-06) pass. The single-fault sweep injects seven fault types in three situations each, and the boat never leaves the fence while driving.

5. Mechanical design

The mechanical design is a parametric CadQuery model: change a parameter and the parts, assembly, mass properties, hydrostatics, renders, STEP and STL files and the design description all rebuild. Hydrostatics are computed by clipping the actual hull mesh at the waterline, so draft, trim, heel and the stability curve come from the real geometry, not a box approximation.

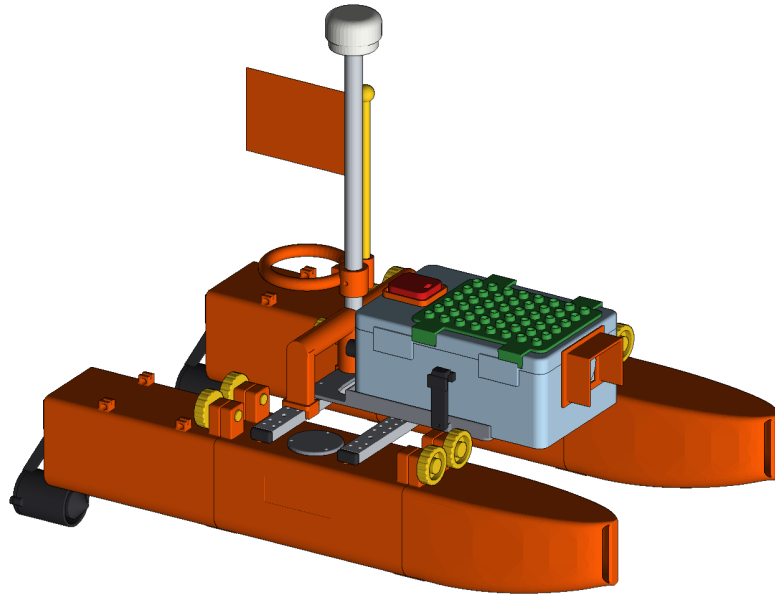


Figure 10. The assembly as modelled: printed ASA hulls in three segments, aluminium crossbeams, the sealed electronics box with its DUPLO deck, the mast and clip-on pods.

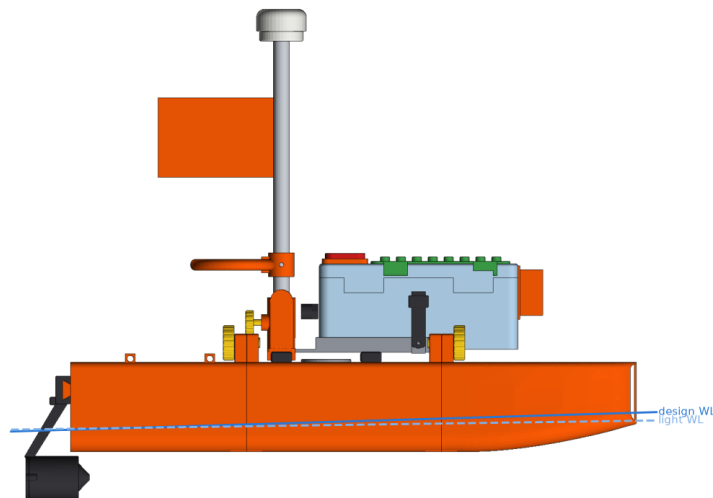


Figure 11. Side view at the computed waterlines: light (dashed) and at design mass with 300 g of payload (34.4 mm draft).

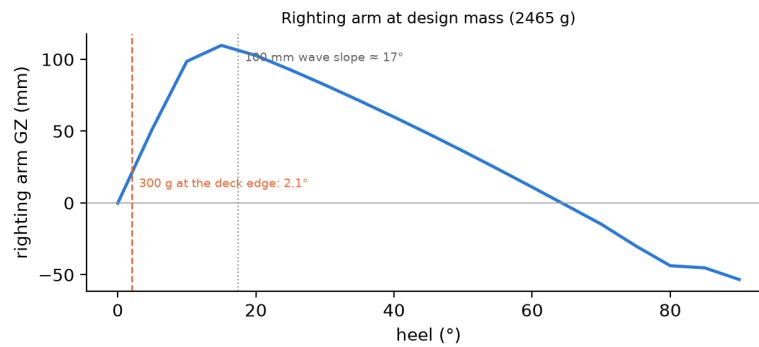


Figure 12. Stability: the righting arm peaks at 110 mm at 15° and stays positive to 65°, far beyond what a pond wave or a child leaning on the deck can do.

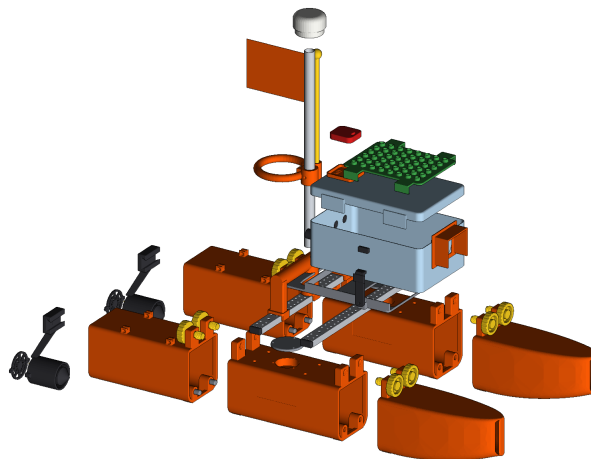


Figure 13. Exploded view: every part is printable on a 200 mm printer or bought off the shelf.

Key results

Check	Result	Limit	
Hull draft at design mass (boat + 300 g on the deck)	34.4 mm at 2465 g	≤ 35	PASS
Least freeboard at design mass	50.7 mm	≥ 50	PASS
Trim (camera level) light / design	-0.8° / -1.7°	$\pm 3^\circ$	PASS
Buoyancy with two segments breached, box flooded	7587 g	$\geq 1.5 \times 2100 = 3150$ g (and $\geq 1.5 \times 2465 = 3697$ g)	PASS
Static heel, 300 g at the deck edge	2.1°	$\leq 10^\circ$	PASS
Stability: max righting arm; angle of vanishing stability	GZ 110 mm at 15°; vanishes at 65°	vanishing $>$ wave slope 17° + edge heel 2°	PASS
Tow by the hoop without capsize (sideways, worst case)	capsizes at 15 N side pull	$>$ side drag at 0.5 m/s ≈ 4.7 N	PASS
Ready-to-sail mass, excluding payload	2165 g	≤ 1800 g	FAIL
Hull shell: material, perimeters, wall	ASA; 3 perimeters; 1.2 mm	PETG/ASA; ≥ 3 ; ≥ 1.6 mm	FAIL
Masthead top above the water (design mass)	415 mm	≈ 450 , total height ≤ 500	FAIL

33 checks pass, 3 need a physical test, 4 are open. The open items are carried by change requests: CR-09 (mass: 2165 g against 1800 g; the options are to accept it, which hydrostatics supports, or to spend effort on lighter hulls), CR-10 (masthead 415 mm above the water against about 450 mm) and CR-11 (1.2 mm hull shell against the 1.6 mm the spec asks, chosen for mass, with the foam core giving the buoyancy).

6. Electrical design

The electrical design is one netlist in code: 120 parts, 70 nets and 114 wires. Schematics, stripboard layouts, the harness wire list and the box layout are all generated from it, and an electrical rule check, a board-copper check and a schematic completeness check run on every build (all zero errors).

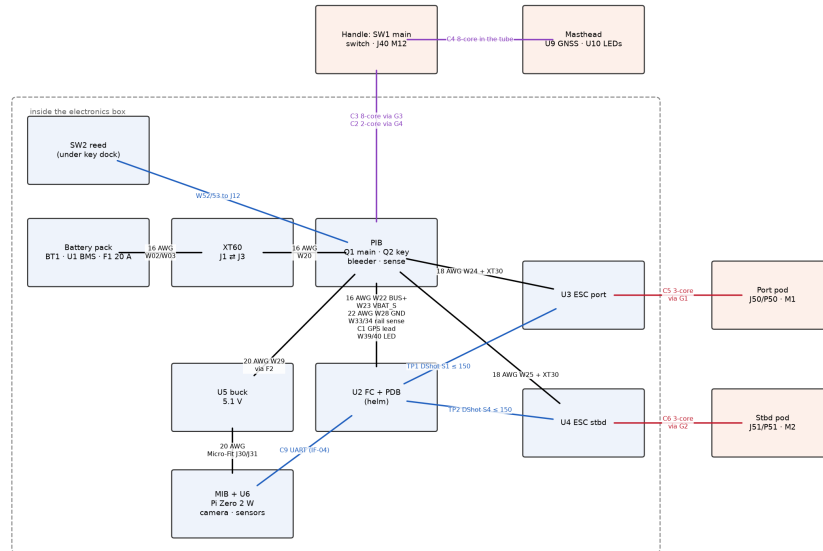


Figure 14. Boards, modules and the harness. Everything outside the box enters through a cable gland.

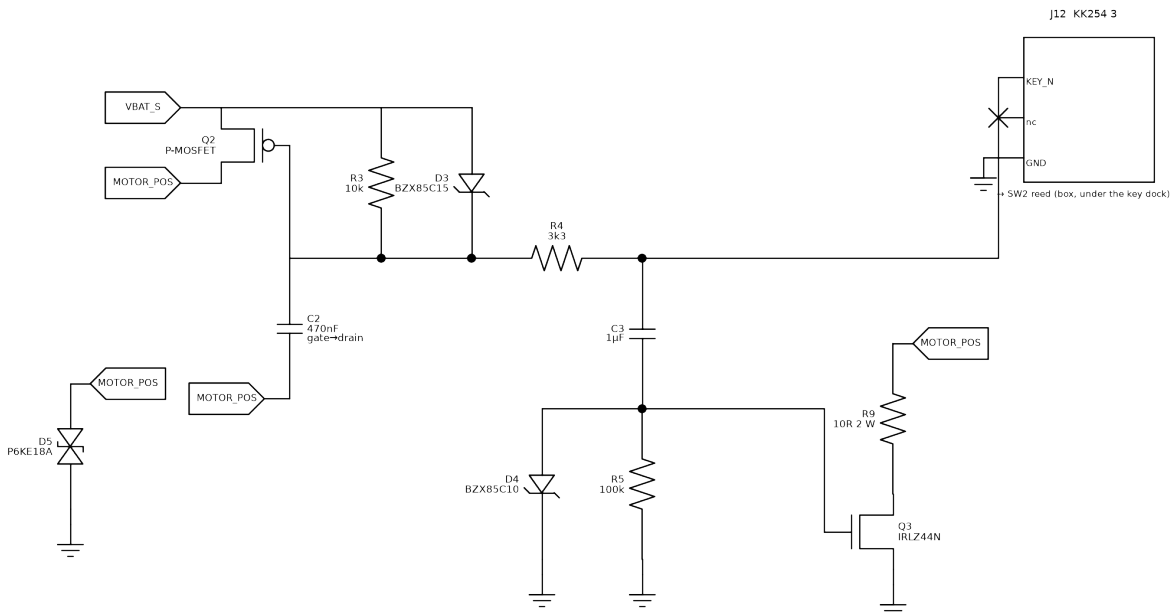


Figure 15. Schematic extract (sheet S2): the magnetic key switch. Q2 is a high-side P-MOSFET held off by R3, so the boat fails safe. C2 (gate to drain) limits the in-rush when the key goes in; when it comes out, C3 pulses Q3 and R9 empties the motor controllers' capacitors.

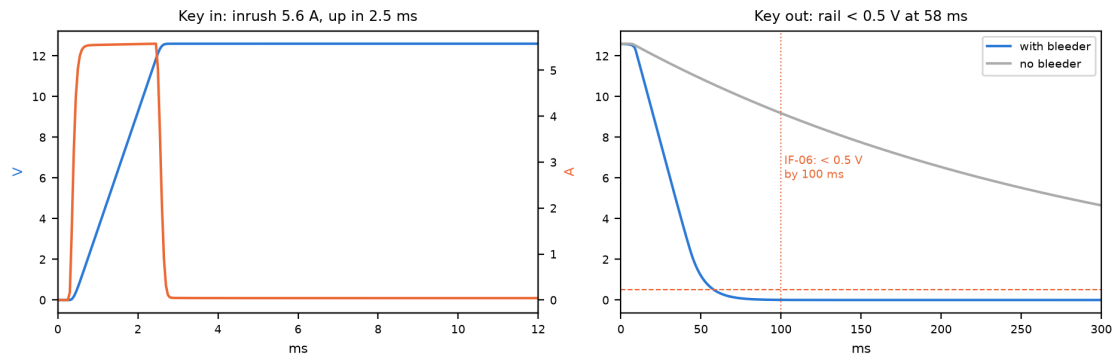


Figure 16. Supporting calculation, simulated: a first design with a simple gate RC gave a 24 A in-rush; the gate-drain capacitor brings it to 5.6 A. Without the bleeder the motors would stay live for nearly a second after the key is pulled; with it, under 100 ms as the interface requires.

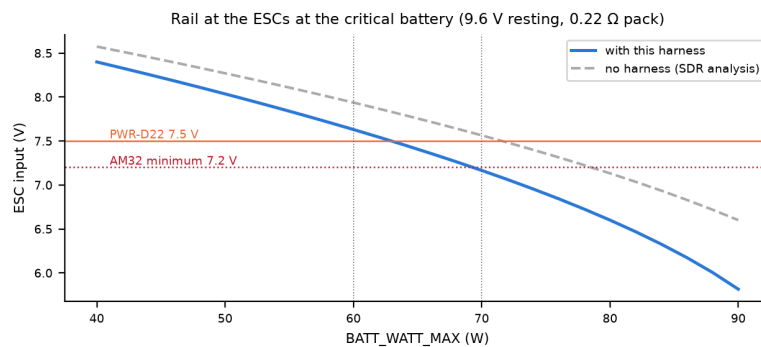


Figure 17. Supporting calculation: voltage at the motor controllers against battery power through the real harness resistance. This is what raised CR-12.

Key results

Check	Result	Limit	
Netlist rule check (every pin placed, nets joined only by boards, mates and wires, drivers, voltages, keying)	0 errors; 120 parts, 70 nets, 114 wires	0 errors	PASS
Stripboard layouts rebuilt from copper, cuts and links, compared with the netlist	0 errors on 3 boards	0 errors	PASS
Key switch: high-side P-MOSFET, soft start, off without the key	inrush 5.6 A, rail up in 2.5 ms	high side; soft start; fails off	PASS
Key out → rail < 0.5 V within 100 ms	58 ms (no bleeder: 956 ms)	≤ 100 ms	PASS
Endurance at cruise, with this design's losses	11.34 W → 114 min on 21.6 Wh	≥ 40 min, margin ≥ 30%	PASS
Box temperature, 30 °C in sun, 60 min	44 °C (3.6 W inside, 7.5 W sun)	≤ 50 °C	PASS
Switch MOSFETs at the power-limited current	T _j 78 °C with a clip-on sink (127 °C without)	< 150 °C	PASS
Every supply works at the rail voltage the power limit allows (critical battery, acceptance-limit cells)	ESC input 7.17 V at 70 W (harness 34 mΩ); 7.5 V needs ≤ 63 W	≥ 7.5 V (ESC rated 7.2 V)	FAIL
Bill of materials within the £185 cap (boat and shore kit)	£246 buying everything; £229 with workshop stock (wire, resistors, stripboard)	≤ £185	FAIL

35 checks pass, 4 need a bench test, 2 are open. CR-12 limits battery power to 60 W so the motor controllers see at least 7.5 V (thrust falls only to 1.85 N); CR-13 records the key-switch circuit change above; CR-14 puts the cost (£229 using workshop stock, £246 buying everything, against £185) to the owner.

Next steps

Step	What	Gate
1	Owner decisions on CR-09 to CR-14 (mass, mast height, shell thickness, power limit, key switch, cost)	CDR
2	Critical Design Review of the MDD and EDD against the baseline	CDR
3	Order parts; print hulls; build the three boards; bench bring-up with the hardware-in-the-loop rigs	TRR
4	Float, stability and pool tests against the verification matrix; the same scenarios as the simulation, now in water	TRR
5	First pond mission with Dad and son: explore, take pictures of the ducks, come back	ORR

What this shows. Strong systems-engineering structure is what makes an AI engineering team trustworthy. Claude produced the work at a pace no small team could, but the value an executive can rely on comes from the scaffolding: traceable requirements, decisions held by a human, documents generated from analysis, an independent review that found real problems, a frozen baseline, and test results that report their gaps.